Temperature-Scaled Knowledge Distillation for Calibrating Quantized Language Models

Jiaming Liu
Columbia University
jl7250@columbia.edu

Leah Li
Columbia University
ql2481@columbia.edu

Haotian Lei
Columbia University
hl3945@columbia.edu

Yanhao Bai
Columbia University
yb2630@columbia.edu

***Abstract*—Reducing large language models to 4-bit precision significantly reduces memory usage and inference latency, but its effect on model calibration—the match between predicted confidence and actual accuracy—has not been well explored. We evaluate calibration across five quantization formats, three model families, and three benchmark tasks, finding that post-training quantization does not reliably improve calibration compared to FP16.**

We further study whether knowledge distillation can recover calibration quality in NF4-quantized models, using LoRA adapters under both single-temperature and asymmetric dual-temperature settings. The effects of KD differ substantially across architectures. For LLaMA-1 7B, calibration improves with higher temperatures but at the cost of accuracy, particularly on factoid tasks when trained on general web text. Mistral 7B becomes unstable at $T \geq 4$, with accuracy collapsing from 82.6% to 55.8% between $T = 2$ and $T = 4$, potentially due to its sharper instruction-tuned output distribution. LLaMA-2 13B behaves most consistently, showing monotonic ECE improvement on PubMedQA (ΔECE $= -0.026$) without sacrificing accuracy. Asymmetric dual-temperature KD partially recovers factoid accuracy for LLaMA-1 on TriviaQA (0.575 to 0.654 at $T_t = 2$, $T_s = 1$), but does not stabilize Mistral. Finally, we find that ECE can be misleading when output distributions collapse toward uniformity, producing artificially low scores that mask degraded predictions; Brier Score and entropy provide a more reliable signal in such cases.

***Index Terms*—knowledge distillation, LLM quantization, calibration, ECE, Brier score, LoRA, NF4, GPTQ, AWQ**

I. Introduction

Deploying large language models at scale requires compression to remain cost-effective. The idea is to reduce costs without affecting downstream systems. A 13B model in FP16 requires about 26 GB of GPU memory; 4-bit quantization reduces this to under 8 GB with minimal accuracy loss. Methods like GPTQ, AWQ, and NF4 are already widely used in practice, yet their effect on calibration is rarely studied, even though calibration matters as much as accuracy in high-stakes settings. In medical QA, legal reasoning, and financial analysis, a model that assigns high confidence to every prediction provides no useful signal for downstream decision-making.

Expected calibration error (ECE) is the standard measure of the gap between accuracy and confidence. It is widely used because it provides a convenient scalar summary of calibration quality. However, ECE can be misleading when output distributions collapse toward uniformity, producing low error scores that do not reflect genuine calibration improvement. In such cases, complementary metrics such as Brier Score and entropy are necessary to distinguish real gains from distributional artifacts.

Knowledge distillation offers a promising approach to improve calibration in quantized models. By training on the teacher's soft outputs rather than hard labels, the student learns a richer uncertainty signal across classes, with temperature controlling the softness of that distribution. Jin et al. [1] showed that using separate temperatures for teacher and student, which they call asymmetric distillation, reduces training instability at high temperatures while preserving the calibration signal. Whether this approach can recover calibration quality in 4-bit quantized LLMs and how sensitive it is to model architecture remains an open question.

In this paper, we apply asymmetric-temperature KD to NF4-quantized LLMs and study three questions: (1) does post-training quantization affect calibration beyond accuracy? (2) which KD temperature works best, and is the answer model-dependent? (3) can ECE reliably reflect calibration quality, or can it give misleading results? We make the following contributions:

- We benchmark calibration across five quantization formats on three models and three tasks, reporting ECE, MCE, Brier Score, and throughput.
- We run a KD temperature sweep under both single and asymmetric dual-temperature settings, identifying three distinct model behaviors including a sharp collapse in Mistral 7B.
- We show that low ECE can reflect degenerate near-uniform predictions rather than genuine calibration improvement, and that Brier Score and entropy are necessary complements.
- We provide practical guidelines for choosing quantization formats and KD temperatures across model families.
- We release a reusable calibration evaluation pipeline covering model loading, confidence extraction, metric computation, and visualization.

II. Literature Review

A. Post-Training Quantization for LLMs

GPTQ [2] reduces reconstruction error through layer-wise second-order weight perturbations, compressing 175B models to 4-bit with less than one percent increase in perplexity. AWQ

TABLE I
APPROXIMATE GPU MEMORY (GB). VALUES INCLUDE MODEL WEIGHTS ONLY.

Model	FP16	GPTQ-8	GPTQ-4	AWQ-4	NF4
LLaMA-1 7B	13.5	6.9	3.9	3.9	4.0
Mistral 7B	14.5	7.4	4.2	4.2	4.3
Llama-2 13B	26.0	13.2	7.3	7.3	7.5

[3] takes a different approach, identifying that weight channels with strong activation influence account for most quantization error and protecting them through rescaling before quantization. QLoRA [4] introduces NF4, a 4-bit format optimized for normally distributed weights, using double quantization to further reduce memory cost. These methods are typically evaluated on accuracy and perplexity; their effect on calibration remains less studied.

B. Calibration of Neural Networks

Guo et al. [5] show that modern neural networks trained with cross-entropy are often overconfident, and that temperature scaling is a simple and effective post-hoc remedy. Kadavath et al. [6] find that LLMs are generally better calibrated on multiple-choice tasks than on open-ended generation, with calibration quality varying substantially across domains. The Brier Score [7] complements ECE as a proper scoring rule that penalizes both overconfidence and underconfidence, making it useful for detecting cases where ECE alone is misleading.

C. Knowledge Distillation for Calibration

Hinton et al. [8] show that the teacher's soft outputs carry dark knowledge, encoding information about inter-class similarities, that improves the student's calibration beyond what hard labels provide. Tang et al. [9] confirm this finding even when the teacher itself is not explicitly calibrated, and Müller et al. [10] show that label smoothing produces similar benefits. Jin et al. [1] propose asymmetric temperature KD, which stabilizes training at high temperatures by decoupling teacher and student temperatures while preserving the calibration signal. We build on this approach and apply it to 4-bit quantized LLMs with LoRA adapters.

III. EXPERIMENTAL SETUP

A. Models and Memory Footprint

We study three model families spanning 7B and 13B parameters:

- **LLaMA-1 7B** [11]: the original Meta LLaMA architecture with RoPE positional encoding.
- **Mistral 7B Instruct v0.2** [12]: grouped-query and sliding-window attention for efficient long-context inference.
- **Llama-2 13B** [13]: second-generation Meta LLaMA at $1.86\times$ the parameter count of the 7B models.

Table I shows GPU memory for each model-quantization combination.

B. Quantization Methods

Each model is evaluated under five precision settings. FP16 serves as the teacher reference; the other four are PTQ variants. GPTQ models use 128-group quantization with activation ordering (`actorder=True`). AWQ models use pre-computed scales from the AWQ calibration procedure. NF4 is loaded via BitsAndBytesConfig with double quantization enabled and FP16 compute dtype.

C. Knowledge Distillation Setup

KD training attaches LoRA adapters ($r = 16$, $\alpha = 32$, target modules: `q/k/v/o_proj`, dropout 0.05) to frozen NF4 weights. Training uses 2,000 examples from C4 [14], filtered to sequences of at least 256 tokens (truncated to 1,024), for 2 epochs with AdamW (lr $= 2\times10^{-4}$, weight decay 0.01), gradient accumulation over 16 steps, cosine LR schedule with 3% warmup, and gradient clipping at 1.0. The KD loss is:

$$\mathcal{L} = \alpha \cdot T_t T_s \cdot \mathrm{KL}\Big(p_{\text{FP16}}^{T_t} \Big\| p_{\text{NF4}}^{T_s}\Big) + (1-\alpha)\cdot \mathrm{CE}(p_{\text{NF4}}, y) \quad (1)$$

where $\alpha = 0.7$ and the $T_t T_s$ prefactor restores gradient scale at high temperatures [1]. We sweep single-temperature ($T_t = T_s = T \in \{1, 2, 4, 8, 12, 16\}$) and six asymmetric dual-temperature combinations ($T_t \in \{2, 3, 4, 5\}$, $T_s \in \{1, 2, 3\}$, $T_t > T_s$). All KD experiments use NF4 as the student format, because NF4 integrates directly with LoRA backpropagation via the QLoRA/bitsandbytes workflow; AWQ and GPTQ do not support this in our implementation.

D. Evaluation Protocol and Metrics

We evaluate on three datasets: **HellaSwag** [15] (4-way commonsense sentence completion, multiple-choice), **TriviaQA** [16] (open-domain factoid QA, greedy generation), and **PubmedQA** [17] (3-way biomedical QA: yes/no/maybe, multiple-choice).

HellaSwag and PubmedQA use log-likelihood scoring over choices (`eval_multiple_choice`), with confidence from the softmax over per-choice log-likelihoods. TriviaQA uses greedy decoding (`eval_generative_qa`), with confidence as the geometric mean of per-token log-probabilities. These two confidence definitions sit on different scales, so TriviaQA ECE values should not be compared directly to those of the other two tasks.

KD evaluations use 1,000 samples per dataset—evaluating every checkpoint on full splits would require prohibitively many GPU-hours given the temperature sweep across three models. PTQ baselines use full splits (HellaSwag: 10,042; TriviaQA: 17,944; PubmedQA: 1,000 from `pqa_labeled` train). Direct PTQ-vs-KD comparisons should account for this sample-size difference. To verify cross-split stability, we re-evaluated NF4 baselines on 1,000-sample subsets; HellaSwag and TriviaQA metrics differed by less than 2pp and 0.012 ECE respectively, confirming that KD comparisons on matched 1,000-sample subsets are reliable.

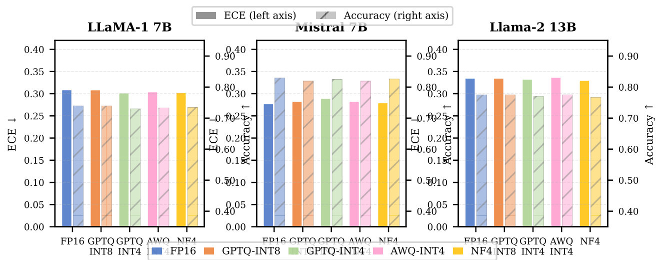

Fig. 1. ECE (solid bars) and accuracy (hatched bars) on HellaSwag. ECE differences among 4-bit methods stay within 0.01 for LLaMA-1 and Llama-2; Mistral 7B shows wider spread on PubmedQA (Table II).

We report ECE (15-bin), MCE (maximum calibration error), Brier Score (proper scoring rule), and Avg. Entropy (distributional sharpness). All experiments ran on a RunPod A100 SXM4-80 GB; KD training takes roughly 2.5 h per (model, temperature) pair.

E. Calibration Evaluation Pipeline

We build a reusable evaluation pipeline that loads each quantized model, runs the task, extracts confidence scores using softmax over log-likelihoods for multiple-choice tasks and token-level geometric mean for generative tasks, and saves both summary metrics and raw tensors. The pipeline computes ECE, MCE, Brier Score, and entropy, and generates reliability diagrams and entropy distribution plots, ensuring consistent comparison across all PTQ and KD configurations without per-model evaluation code.

IV. PTQ Calibration Results

A. Accuracy and ECE Across Methods

Table II reports accuracy and ECE across all five formats. Fig. 1 visualizes HellaSwag results.

Quantization causes some accuracy drop, but the effect varies by task. On LLaMA-1 7B, NF4 loses about 0.5 points on HellaSwag but 2.4 points on TriviaQA compared to FP16, and GPTQ-INT4 drops 3.6 points on TriviaQA against only 0.9 points on HellaSwag, suggesting that factoid retrieval tasks are more sensitive to quantization error than commonsense multiple-choice tasks.

For calibration, no quantization method consistently outperforms FP16 across all models and datasets. One notable exception is Mistral 7B on PubMedQA, where GPTQ-INT4 reduces ECE from 0.229 to 0.077 and AWQ-INT4 reduces it further to 0.067. However, this does not reflect genuine calibration improvement; the lower ECE likely stems from smoother logits or reduced confidence rather than better probability estimates, underscoring the need to examine Brier Score and entropy alongside ECE.

B. Throughput Analysis

Fig. 2 shows inference throughput. GPTQ-INT4 achieves the best speedup—3.4× for LLaMA-1 7B, 1.1× for Llama-2 13B. NF4 throughput (727–908 tok/s) runs below AWQ-INT4 (672–1,221 tok/s) for LLaMA-1, a result of the bitsandbytes dequantization path that varies across hardware and CUDA versions. GPTQ-INT8 shows inconsistent speedup: 3.2× speedup on LLaMA-1 but only 0.21× on Llama-2 13B, likely due to memory bandwidth saturation at the larger model size.

The throughput-ECE tradeoff is favorable for both GPTQ-INT4 and NF4: near-FP16 ECE at 4× lower memory and 1.1–3.4× higher throughput. NF4's lower raw throughput is offset by its compatibility with LoRA adapters for post-training calibration.

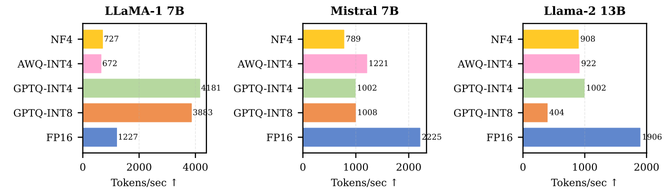

Fig. 2. Inference throughput (tok/s, HellaSwag, A100 SXM4-80GB). GPTQ-INT4 reaches 3.4× FP16 throughput for LLaMA-1 7B. NF4 is slower due to bitsandbytes dequantization on every forward pass.

V. Knowledge Distillation Results

A. Overview: ECE and Accuracy vs. Temperature

Figures 3 and 4 show how ECE and accuracy evolve across the temperature sweep, revealing three distinct behavioral patterns. LLaMA-1 7B is largely insensitive to temperature within the KD sweep, Mistral 7B shows a sharp break around ($T = 4$), and Llama-2 13B behaves most smoothly, with ECE on PubMedQA improving steadily as temperature increases.

B. LLaMA-1 7B: Temperature-Stable with Accuracy Trade-offs

Table III gives the full results for LLaMA-1 7B. Within the KD runs, HellaSwag accuracy barely changes (0.695–0.701). Compared to the NF4 baseline, however, performance is clearly worse: HellaSwag drops by about 3–4 points, and TriviaQA falls by 8–9 points (0.662 → 0.575–0.582). This suggests that KD on C4 does not provide a strong enough signal for knowledge-heavy tasks like TriviaQA. While ECE on TriviaQA improves (0.018–0.022 vs. NF4), this comes at the cost of accuracy. On HellaSwag, ECE also improves (ΔECE ≈ -0.025 to -0.032), but the Brier Score barely changes (≤ 0.002). This suggests the ECE gain reflects reduced confidence as accuracy drops, rather than genuinely improved probability estimates.

The clearest positive result for LLaMA-1 is on PubmedQA at ($T = 1$). Here, ECE drops from 0.247 to 0.218 and Brier Score from 0.325 to 0.310, while accuracy stays essentially the same (0.530 vs. 0.532 baseline). This is the only configuration where calibration improves without an accuracy cost.

C. Mistral 7B: Sharp Temperature-Induced Collapse

Mistral 7B is where temperature really starts to matter (Table IV). At $T = 1$, the HellaSwag accuracy is 74.6%, and it

TABLE II
PTQ Accuracy and ECE Across Models, Quantization Methods, and Datasets. Best ECE per model-dataset in **bold**.

Model	**Method**	**HellaSwag** Acc ↑	ECE ↓	**TriviaQA** Acc ↑	ECE ↓	**PubmedQA** Acc ↑	ECE ↓
LLaMA-1 7B	FP16	0.7395	0.3086	0.6854	0.0837	0.5360	0.2392
	GPTQ-INT8	0.7394	0.3085	0.6860	0.0823	0.5350	0.2310
	GPTQ-INT4	0.7300	0.3015	0.6494	0.0763	0.5390	0.2402
	AWQ-INT4	0.7328	0.3040	0.6752	**0.0707**	0.5390	0.2552
	NF4	0.7342	**0.3021**	0.6616	0.0750	0.5320	0.2469
Mistral 7B	FP16	**0.8293**	0.2771	**0.7005**	0.1427	0.4910	0.2293
	GPTQ-INT8	0.8191	0.2827	0.6778	0.1550	0.4610	0.1033
	GPTQ-INT4	0.8246	0.2893	0.6807	0.1525	**0.4930**	**0.0766**
	AWQ-INT4	0.8192	0.2825	0.6854	0.1545	0.4920	0.0667
	NF4	0.8259	**0.2793**	0.6870	0.1581	0.4810	0.2361
Llama-2 13B	FP16	0.7750	0.3350	**0.7670**	0.1667	0.4700	0.2293
	GPTQ-INT8	0.7750	0.3349	0.7630	0.1633	0.4730	0.2284
	GPTQ-INT4	0.7690	0.3326	0.7430	**0.1549**	0.4730	**0.2197**
	AWQ-INT4	**0.7750**	0.3368	0.7520	0.1647	0.4710	0.2415
	NF4	0.7670	**0.3298**	0.7540	0.1775	0.4670	0.2400

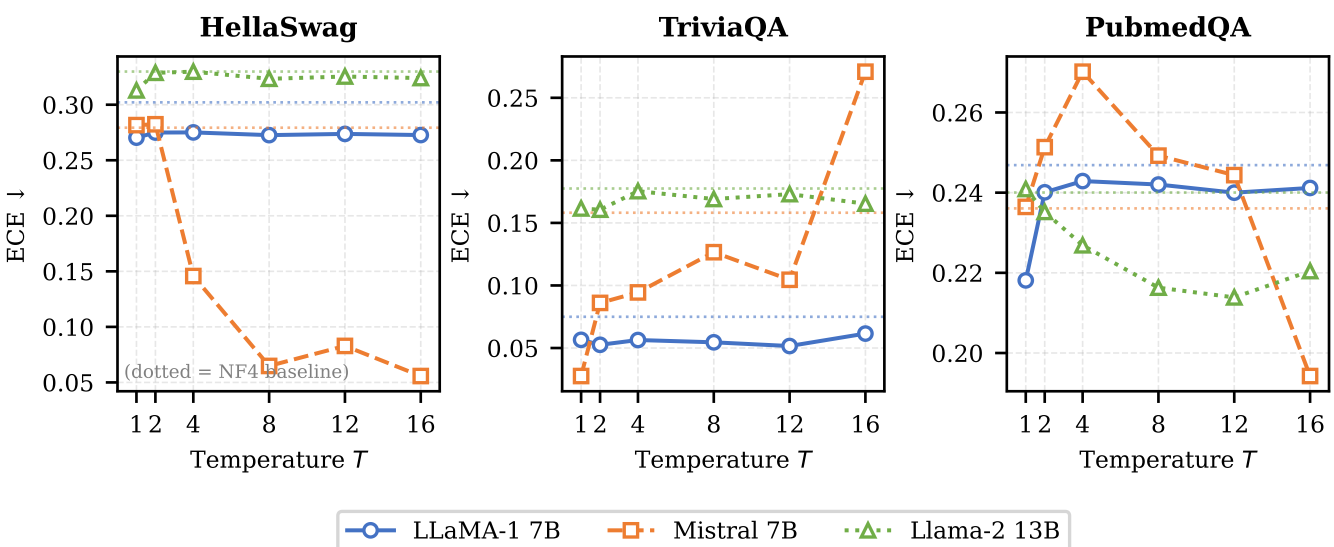

Fig. 3. ECE vs. KD temperature for all three models on HellaSwag (left), TriviaQA (center), and PubmedQA (right). Dotted lines indicate NF4 baseline ECE. Mistral's ECE drop at $T \geq 4$ is an artifact of distribution collapse (§V-C). Llama-2's monotonic decrease on PubmedQA is genuine (§V-D).

TABLE III
KD Temperature Sweep — LLaMA-1 7B (1,000 eval samples). NF4 baseline in top row.

T	HellaSwag Acc	ECE	Brier	TriviaQA Acc	ECE	Brier	PubmedQA Acc	ECE	Brier
NF4	.7342	.3021	.2701	.6616	.0750	.2280	.5320	.2469	.3246
1	.6950	.2703	.2687	.5750	.0565	.2411	.5300	.2181	.3099
2	.7010	.2749	.2699	.5750	.0526	.2404	.5360	.2401	.3193
4	.7000	.2751	.2699	.5790	.0564	.2433	.5370	.2429	.3206
8	.6980	.2726	.2694	.5820	.0546	.2437	.5360	.2420	.3203
12	.6990	.2737	.2697	.5820	.0516	.2422	.5360	.2400	.3194
16	.6980	.2727	.2694	.5810	.0615	.2429	.5360	.2412	.3196

TABLE IV
KD Temperature Sweep — Mistral 7B. Collapse at $T = 4$.

T	HellaSwag Acc	ECE	Brier	TriviaQA Acc	ECE	Brier	PubmedQA Acc	ECE	Brier
NF4	.8259	.2793	.2012	.6870	.1581	.2298	.4810	.2361	.3043
1	.7460	.2816	.2506	.6350	.0276	.2258	.4940	.2365	.3071
2	.7820	.2824	.2304	.6660	.0861	.2150	.4650	.2513	.3174
4	.5580	.1458	.2557	.4980	.0944	.2555	.4390	.2702	.3377
8	.4560	.0648	.2451	.3950	.1266	.2431	.4340	.2492	.3304
12	.4710	.0829	.2490	.4300	.1045	.2573	.4180	.2444	.3205
16	.4340	.0558	.2429	.3010	.2709	.2786	.4260	.1943	.3017

actually improves to 78.2% at $(T = 2)$. This does not last. By $(T = 4)$, accuracy drops sharply to 55.8%, and keeps falling to 43.4% at $(T = 16)$.

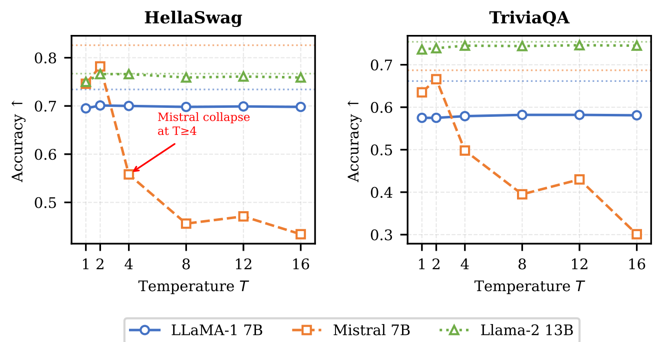

Fig. 4. Accuracy vs. KD temperature. Mistral 7B collapses from 82.6% to 55.8% between $T = 2$ and $T = 4$. LLaMA-1 7B is stable across temperatures but sits 3–4 pp below NF4 on HellaSwag and 8–9 pp below on TriviaQA. Llama-2 13B tracks its NF4 baseline most closely.

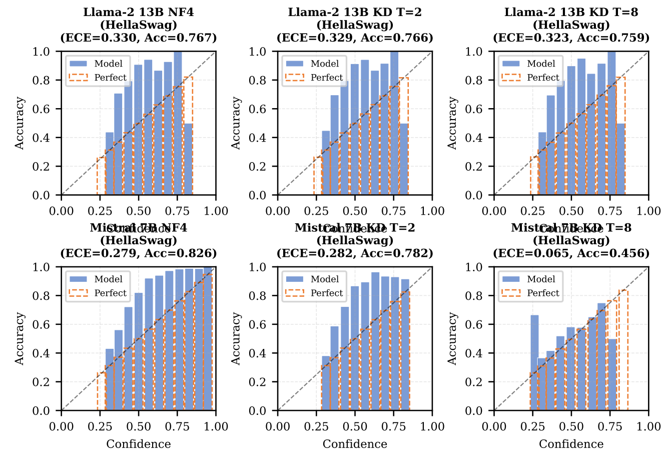

Fig. 5. Reliability diagrams (HellaSwag, 1,000 samples) at NF4 baseline, KD $T = 2$, and KD $T = 8$. Top: Llama-2 13B. Bottom: Mistral 7B. Mistral at $T = 8$ shows near-uniform confidence (degenerate); Llama-2 remains structured.

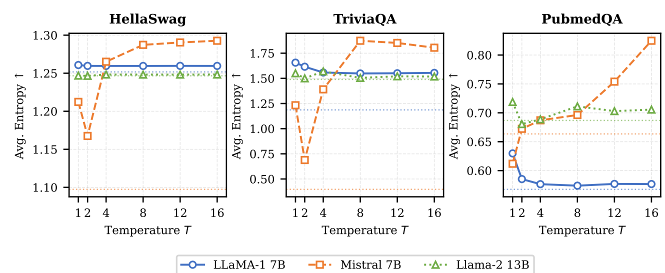

Fig. 6. Average entropy vs. KD temperature. Mistral's sharp rise at $T \geq 4$ signals distribution collapse before ECE turns misleading.

The improvement in ECE at higher temperatures ($T \geq 8$) is a bit deceptive. Even though ECE goes down, the Brier Score moves in the opposite direction, increasing from 0.201 (NF4 baseline) to 0.246 at ($T = 8$). This divergence confirms the ECE improvement does not reflect a real calibration gain.

Figure 5 makes this clearer. At ($T = 8$), the model is basically assigning the same probability to all four HellaSwag options—around 25% each. So its confidence ends up lining up with random guessing accuracy. This alignment with random-guessing accuracy produces a misleadingly low ECE while the model has effectively stopped making meaningful distinctions.

Figure 6 shows a similar pattern from another angle. For Mistral, average entropy starts to climb once ($T \geq 4$), going from 1.17 bits at ($T = 2$) to 1.29 bits at ($T = 16$). LLaMA-1 7B and Llama-2 13B, by contrast, stay almost unchanged at around 1.25 bits. This entropy increase while ECE decreases is a reliable warning sign that the model is drifting toward uniform predictions rather than improving its probability estimates. We hypothesize that Mistral's base distribution is already quite sharp, so increasing temperature flattens it too much. Without access to teacher logits, though, this remains speculative.

D. Llama-2 13B: Consistent Calibration Gains

Llama-2 13B behaves much more consistently (Table V). On HellaSwag, accuracy at ($T = 2$) (76.6%) is essentially the same as the NF4 baseline (76.7%). TriviaQA reaches 74.5% at ($T = 4$–8), which is still close to the baseline of 75.4%. On PubmedQA, ECE decreases steadily as temperature increases: $0.240 \rightarrow 0.235 \rightarrow 0.227 \rightarrow 0.216 \rightarrow 0.214$ (from NF4 to ($T = 12$)), giving a total drop of -0.026. The Brier Score also improves ($0.310 \rightarrow 0.299$ at ($T = 8$)), which suggests this is a real calibration improvement rather than an artifact of flattening. Entropy stays almost constant (around 1.248 bits), so the model does not lose its ability to separate choices.

E. Asymmetric Dual-Temperature KD

We also tried six dual-temperature settings, with teacher temperature (T_t) higher than student temperature (T_s). Table VI compares these against the best single-temperature results.

The most significant effect of dual-temperature training appears for LLaMA-1 on TriviaQA. With a single temperature of ($T = 2$), accuracy drops to 0.575 (from 0.662 at the NF4 baseline). Using a dual setting of ($T_t = 2$), ($T_s = 1$), accuracy comes back up to 0.654, almost closing the gap. In this case, the teacher still provides softer targets, but the student does not fully match an overly flat distribution. This appears to help retain factual information. The trade-off is much worse ECE (0.426 vs. 0.053), since the student outputs are sharper. A more balanced option is ($T_t = 4$), ($T_s = 3$), which keeps accuracy similar (0.650) while bringing ECE down to 0.077, with comparable HellaSwag results.

For Llama-2 13B, dual-temperature training looks very similar to the single-temperature case, with no clear advantage either way.

Mistral again fails: even the mildest setting Tt2_Ts1 collapses HellaSwag accuracy to 0.515, far below single $T = 2$ (0.782). This suggests Mistral's instability is not just about

TABLE V
KD TEMPERATURE SWEEP — LLAMA-2 13B. PUBMEDQA ECE IMPROVES MONOTONICALLY.

T	HellaSwag Acc	HellaSwag ECE	HellaSwag Brier	TriviaQA Acc	TriviaQA ECE	TriviaQA Brier	PubmedQA Acc	PubmedQA ECE	PubmedQA Brier
NF4	.7670	.3298	.2725	.7540	.1775	.2241	.4670	.2400	.3097
1	.7500	.3129	.2684	.7360	.1615	.2263	.4350	.2409	.3196
2	.7660	.3287	.2717	.7390	.1605	.2256	.4750	.2352	.3071
4	.7660	.3297	.2730	.7450	.1754	.2253	.4740	.2269	.3043
8	.7590	.3233	.2709	.7440	.1691	.2230	.4750	.2163	.2991
12	.7610	.3254	.2717	.7460	.1730	.2230	.4800	.2139	.3008
16	.7590	.3240	.2711	.7450	.1654	.2196	.4720	.2204	.2998

TABLE VI
DUAL-TEMPERATURE KD VS. SINGLE-TEMPERATURE (BEST PER MODEL). **BOLD** = BEST TRIVIAQA ACCURACY PER MODEL.

Model	Setting	HellaSwag Acc	HellaSwag ECE	TriviaQA Acc	TriviaQA ECE	PubmedQA Acc	PubmedQA ECE
LLaMA-1	Single $T=2$	.701	.275	.575	.053	.536	.240
	Tt2_Ts1	.710	.314	**.654**	.426	.513	.094
	Tt4_Ts3	.706	.289	.650	.077	.531	.187
Llama-2	Single $T=2$	.766	.329	.739	.161	.475	.235
	Tt4_Ts3	.765	.336	**.738**	.197	.481	.204
Mistral	Single $T=2$	.782	.282	.666	.086	.465	.251
	Tt2_Ts1	.515	.134	.514	.199	.459	.124

teacher softness—even decoupling the student temperature does not stabilize training.

VI. DISCUSSION

The three models behave quite differently under KD: $T = 4$ remains stable for Llama-2 13B but triggers collapse in Mistral 7B, a divergence visible in the entropy plots from that point onward. We hypothesize that Mistral's sharper base distribution after instruction tuning causes increasing temperature to flatten its outputs too quickly, though confirming this would require access to teacher logits. In practice, a small probe of 100 samples is usually sufficient to detect this trend; if accuracy drops by more than 5% relative to NF4, the temperature setting should be rejected before running a full KD job.

A. ECE Alone is Not Enough

At T = 16, Mistral reaches an ECE of 0.056 on HellaSwag, which appears substantially better than FP16 (0.277), yet accuracy collapses to 43.4% from 82.9%. The model is assigning nearly equal probability to all four choices, aligning its confidence with random-guessing accuracy and producing a misleadingly low ECE. The Brier Score reveals the true picture: it increases to 0.243 from 0.197 under FP16, indicating worse overall performance. More generally, when ECE decreases while accuracy drops and entropy rises, the model is drifting toward uniform predictions rather than improving calibration. Brier Score and entropy together make this pathology straightforward to detect.

B. KD Effectiveness Depends on Task and Corpus

TriviaQA behaves differently from HellaSwag and PubMedQA because its confidence signal is defined differently: TriviaQA uses the geometric mean of per-token log-probabilities under greedy decoding, while the other two tasks use choice-level softmax. These two signal types do not respond to soft-target training in the same way, which partly explains why TriviaQA ECE stays low across models and temperatures while accuracy degrades.

Two distinct effects contribute to TriviaQA's accuracy drop under KD, and the current experimental design cannot isolate them. First, C4 lacks the factual density needed for knowledge-heavy retrieval tasks, so any C4-trained KD would reduce factoid recall regardless of temperature. Second, higher temperatures additionally smooth output distributions in a way that compounds the accuracy loss. On LLaMA-1, ECE on TriviaQA decreases by 0.018 to 0.022 while accuracy drops by 8 to 9 points; dual-temperature KD recovers part of this loss but does not eliminate the trade-off. Task-aligned training data may be necessary for knowledge-heavy tasks, though this hypothesis was not tested here.

C. Practical Takeaways

NF4 is the preferred format when post-training calibration via LoRA is planned; GPTQ-INT4 is the better choice for inference-only deployments where throughput matters. For KD temperature, T = 2 is a reliable starting point across all three model families studied here. Asymmetric dual-temperature with Tt = 4, Ts = 3 offers a more balanced ECE-accuracy trade-off on factoid tasks, though it does not fully close the accuracy gap. When accuracy drops by more than 5% relative to the NF4 baseline, the temperature setting should be discarded. The most stable improvement in this study is Llama-2 13B on PubMedQA ($\Delta ECE = -0.026$, $\Delta Brier = -0.009$) without accuracy loss; outside this case, KD effects vary substantially by model and task. ECE should never be interpreted without Brier Score and entropy, as low ECE can reflect degenerate uniform predictions rather than genuine calibration improvement.

D. Limitations

The PTQ baselines use full evaluation splits while KD runs use 1,000-sample subsets; cross-split stability was verified for HellaSwag and TriviaQA but the comparison is not perfectly matched. KD training on C4 rather than task-specific data likely accounts for the TriviaQA accuracy degradation, and task-aligned corpora may yield more stable results. The LoRA rank is fixed at r = 16; higher ranks may transfer calibration information more effectively but were not explored. Quantization-aware training was considered as a stretch goal but not implemented due to compute constraints. Finally, ECE values are not directly comparable across datasets because HellaSwag and PubMedQA use choice-level softmax while TriviaQA uses token-level geometric means, and all experiments use a single random seed, so ECE differences below 0.01 should not be over-interpreted.

VII. Conclusion

We studied calibration across five quantization formats and three model families, applying temperature-scaled KD under both single and asymmetric dual-temperature settings. Our central finding is that PTQ does not reliably improve calibration over FP16; the more consequential factor is how each model responds to KD.

ECE alone is an unreliable calibration indicator. In several runs, particularly Mistral 7B at high temperatures, a low ECE reflects near-uniform output distributions rather than genuine calibration improvement. Brier score and entropy are necessary complements for distinguishing real gains from distributional collapse.

The three models exhibit distinct behaviors under KD. LLaMA-1 7B remains stable across temperatures but ECE improvements are accompanied by accuracy drops, most severely on TriviaQA when trained on general web text. Mistral 7B collapses sharply beyond $T = 2$, a failure that dual-temperature training does not resolve, which we attribute to its sharper instruction-tuned base distribution. Llama-2 13B is the most reliable, reducing ECE monotonically on PubmedQA with a total drop of 0.026 while preserving accuracy.

Asymmetric dual-temperature KD offers a partial remedy for accuracy degradation on factoid tasks: for LLaMA-1 on TriviaQA, accuracy recovers from 0.575 to 0.654 under $T_t = 2, T_s = 1$, though the ECE-accuracy trade-off is not fully eliminated. The most stable result across this study remains Llama-2 13B on PubMedQA, where calibration improves without accuracy cost. Future work should examine task-aligned KD training data, higher LoRA ranks, and teacher logit analysis to test the distribution-sharpness hypothesis for Mistral.

AI Tool Use Disclosure

AI assistance (Claude, Anthropic) was used in two capacities during this project. For the final report, AI tools were used to polish prose and improve writing clarity on drafts written by the authors; all technical content, experimental results, and analysis were produced independently. For implementation, AI tools were used for clarification of library APIs and debugging assistance. All profiling interpretations, performance reasoning, and analytical conclusions are the authors' own work, consistent with the HPML AI Use Policy.

References

[1] X. Jin *et al.*, "Uniformly stable knowledge distillation with asymmetric temperature," in *Advances in Neural Information Processing Systems*, 2022.

[2] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, "GPTQ: Accurate post-training quantization for generative pre-trained transformers," in *Proceedings of the International Conference on Learning Representations*, 2023.

[3] J. Lin, J. Tang, H. Tang, S. Yang, X. Dang, and S. Han, "AWQ: Activation-aware weight quantization for LLM compression and acceleration," in *Proceedings of Machine Learning and Systems*, 2024.

[4] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Advances in Neural Information Processing Systems*, 2023.

[5] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the International Conference on Machine Learning*, 2017, pp. 1321–1330.

[6] S. Kadavath *et al.*, "Language models (mostly) know what they know," *arXiv preprint arXiv:2207.05221*, 2022.

[7] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950.

[8] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," in *NeurIPS Workshop on Deep Learning*, 2015.

[9] J. Tang, R. Shivanna, Z. Zhao, D. Lin, A. Singh, E. H. Chi, and S. Jain, "Understanding and improving knowledge distillation," *arXiv preprint arXiv:2002.03532*, 2020.

[10] R. Müller, S. Kornblith, and G. Hinton, "When does label smoothing help?" in *Advances in Neural Information Processing Systems*, 2019, pp. 4694–4703.

[11] H. Touvron *et al.*, "LLaMA: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, 2023.

[12] A. Q. Jiang *et al.*, "Mistral 7B," *arXiv preprint arXiv:2310.06825*, 2023.

[13] H. Touvron *et al.*, "Llama 2: Open foundation and fine-tuned chat models," *arXiv preprint arXiv:2307.09288*, 2023.

[14] C. Raffel *et al.*, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.

[15] R. Zellers, A. Holtzman, Y. Bisk, A. Farhadi, and Y. Choi, "HellaSwag: Can a machine really finish your sentence?" in *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2019.

[16] M. Joshi, E. Choi, D. S. Weld, and L. Zettlemoyer, "TriviaQA: A reading comprehension dataset over wikipedia and the web," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2017.

[17] Q. Jin, B. Dhingra, Z. Liu, W. W. Cohen, and X. Lu, "PubMedQA: A biomedical research question answering dataset," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2019.